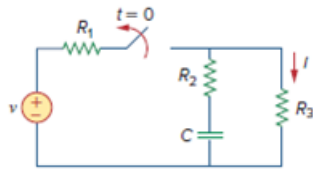


Problem

Determine v and i for $t > 0$ in the circuit of Fig. 8.28. (See comments Practice Problem 8.9 about current sources in Practice Prob. 7.5.)

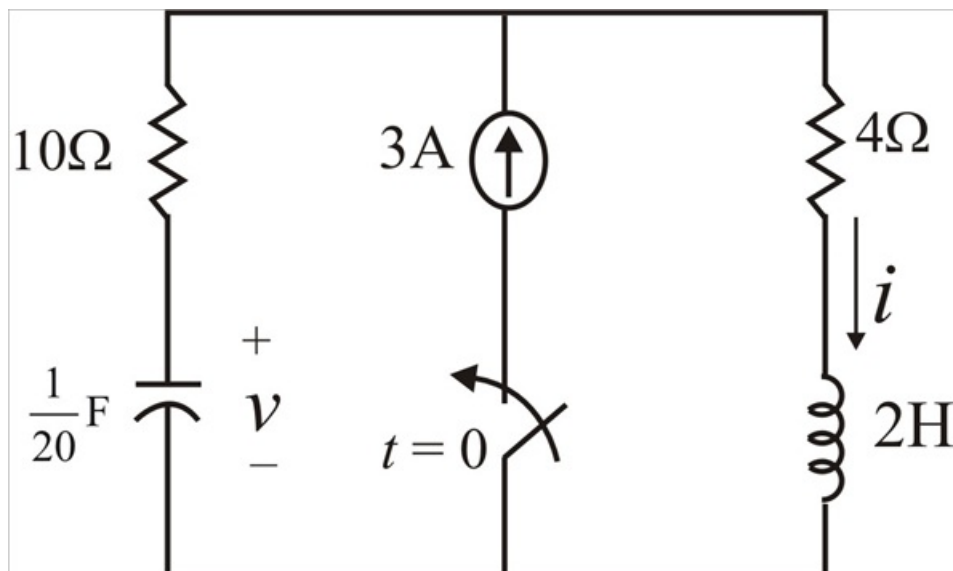
Fig. 8.28



Step-by-step solution

Step 1 of 19

The given circuit is



Step 2 of 19

For $t < 0$, source is not connected to the circuit therefore the initial capacitor voltage and inductor current are zero.

$$\text{i.e., } v(0^-) = 0\text{V}$$

$$i(0^-) = 0\text{A}$$

Since the capacitor voltage and inductor current cannot change instantaneously,

$$v(0^+) = v(0^-)$$

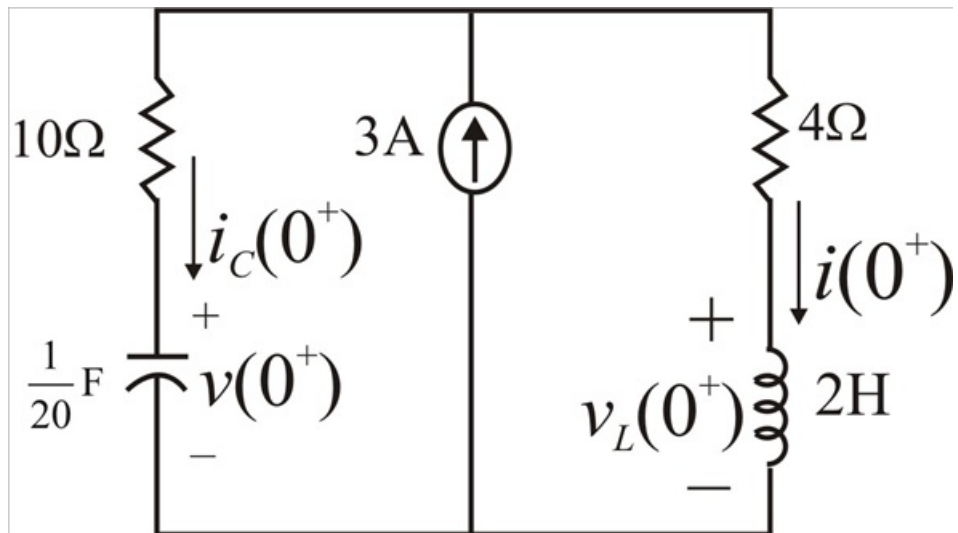
$$v(0^+) = 0\text{V}$$

$$\text{And } i(0^+) = i(0^-)$$

$$i(0^+) = 0\text{A}$$

Step 3 of 19

At $t = 0^+$, the switch is closed therefore the current source is connected to the circuit as shown



Step 4 of 19

Since the inductor current, $i(0^+) = 0\text{A}$

The current 3A flows through the 10Ω resistor and capacitor

$$i_c(0^+) = 3\text{A}$$

The current through the capacitor is,

$$i_c = C \frac{dv}{dt}$$

$$\frac{dv(0^+)}{dt} = \frac{i_c(0^+)}{C}$$

Step 5 of 19

$$\frac{dv(0^+)}{dt} = \frac{3}{\left(\frac{1}{20}\right)}$$

$$\frac{dv(0^+)}{dt} = 60 \text{ V/s}$$

Similarly the voltage across the 10Ω is

$$v_{10\Omega} = 3(10)$$

$$v_{10\Omega} = 30\text{V}$$

Step 6 of 19

The voltage across the 4Ω is

$$v_{4\Omega} = 4i(0^+)$$

$$v_{4\Omega} = 4(0)$$

$$v_{4\Omega} = 0\text{V}$$

Step 7 of 19

Writing KVL in the outer loop

$$-v(0^+) - v_{10\Omega} + v_{4\Omega} + v_L(0^+) = 0$$

$$v_L(0^+) = v(0^+) + v_{10\Omega} - v_{4\Omega}$$

$$v_L(0^+) = 0 + 30 - 0$$

$$v_L(0^+) = 30 \text{ V}$$

Step 8 of 19

The voltage across the inductor is

$$v_L = L \frac{di}{dt}$$

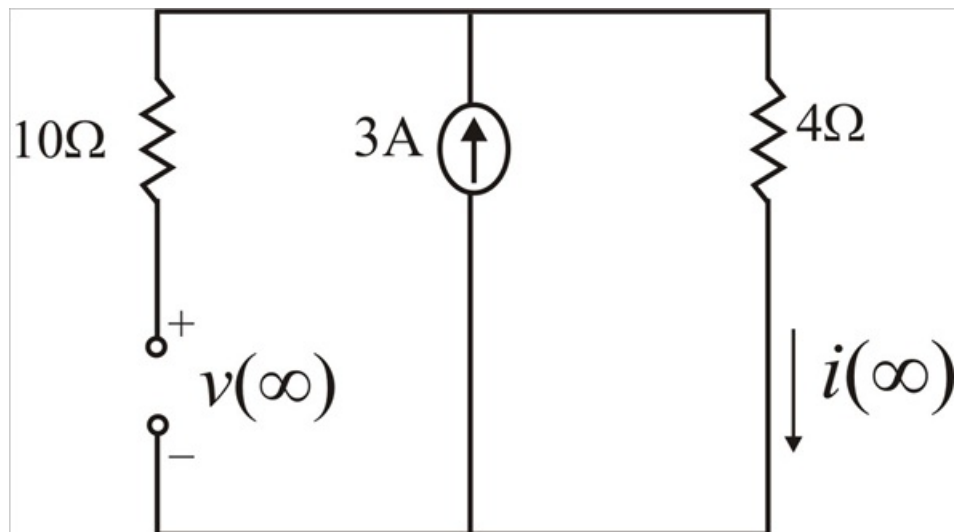
$$\frac{di(0^+)}{dt} = \frac{v_L(0^+)}{L}$$

$$\frac{di(0^+)}{dt} = \frac{30}{2}$$

$$\frac{di(0^+)}{dt} = 15 \text{ A/s}$$

Step 9 of 19

The final values are obtained when the inductor is replaced by a short circuit and the capacitor by an open circuit,



Step 10 of 19

From above circuit, we obtain

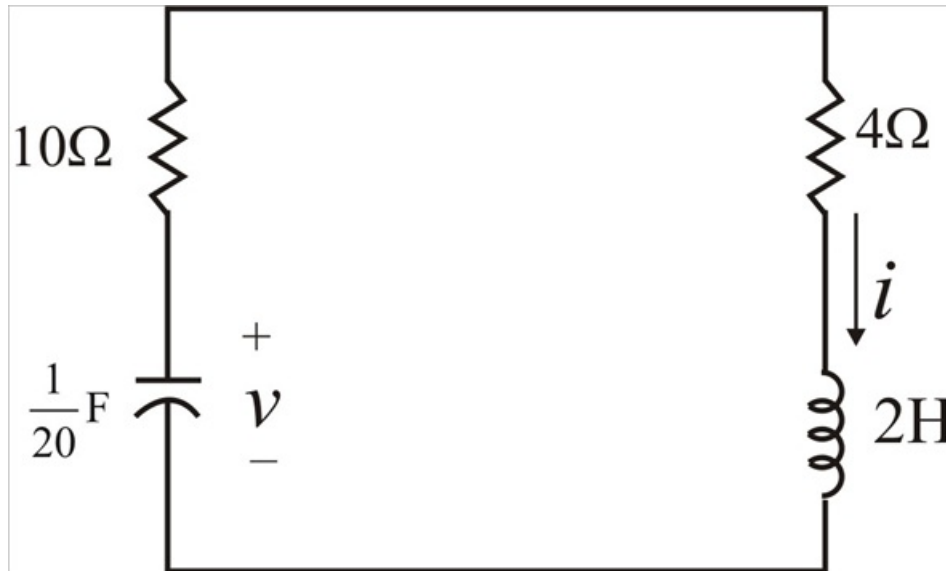
$$i(\infty) = 3\text{A}$$

$$v(\infty) = 4(3)$$

$$v(\infty) = 12 \text{ V}$$

Step 11 of 19

Now we obtain the form of the transient response for $t > 0$. By turning off the 3A current source, we have the circuit as shown



Step 12 of 19

It is a series RLC circuit

$$\alpha = \frac{R}{2L}$$

$$\alpha = \frac{10+4}{2 \times 2}$$

$$\alpha = 3.5$$

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$$\omega_0 = \frac{1}{\sqrt{2 \left(\frac{1}{20} \right)}}$$

$$\omega_0^2 = 10$$

Step 13 of 19

Since $\alpha^2 > \omega_0^2$, the response is over damped and the solution is,

$$v(t) = v_{ss} + Ae^{s_1 t} + Be^{s_2 t}$$

Where the roots are,

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$

$$s_{1,2} = -3.5 \pm \sqrt{12.25 - 10}$$

$$s_{1,2} = -3.5 \pm 1.5$$

$$s_1 = -2$$

$$s_2 = -5$$

Step 14 of 19

And v_{ss} is the steady state response

$$v(\infty) = v_{ss}$$

$$v_{ss} = 12\text{V}$$

$$v(t) = 12 + Ae^{-2t} + Be^{-5t} \dots\dots\dots (1)$$

Using initial conditions

$$v(0) = 12 + A + B$$

$$0 = 12 + A + B$$

$$A + B = -12$$

$$A = -B - 12 \dots\dots\dots (2)$$

Step 15 of 19

Taking the derivative of $v(t)$ in equation (1), we get

$$\frac{dv(t)}{dt} = -2Ae^{-2t} - 5Be^{-5t}$$

At $t = 0^+$,

$$\frac{dv(0^+)}{dt} = -2A - 5B$$

$$60 = -2A - 5B$$

$$2A + 5B = -60 \dots\dots\dots (3)$$

Step 16 of 19

Substituting A from equation (2) into (3), we get

$$2(-B - 12) + 5B = -60$$

$$B = -12$$

From equation (2),

$$A = -(-12) - 12$$

$$A = 0$$

Hence the final solution is,

$$v(t) = 12 - 12e^{-5t}$$

$$\therefore \boxed{v(t) = 12(1 - e^{-5t}) \text{ V}}$$

The current through inductor is

$$i(t) = I_{ss} + Ae^{-2t} + Be^{-5t}$$

Where $I_{ss} = i(\infty)$

Step 17 of 19

$$I_{ss} = 3 \text{ A}$$

$$i(t) = 3 + Ae^{-2t} + Be^{-5t} \dots\dots\dots (4)$$

At $t=0$,

$$i(0) = 3 + A + B$$

$$0 = 3 + A + B \quad A = -B - 3 \dots\dots\dots (5)$$

Step 18 of 19

Taking the derivative of $i(t)$ in equation (4), we get

$$\frac{di(t)}{dt} = -2Ae^{-2t} - 5Be^{-5t}$$

At $t=0^+$,

$$\frac{di(0^+)}{dt} = -2A - 5B$$

$$15 = -2A - 5B$$

$$2A + 5B = -15 \dots\dots\dots (6)$$

Step 19 of 19

Substituting A from equation (5) into (6), we get

$$2(-B - 3) + 5B = -15$$

$$B = -3$$

From equation (5),

$$A = -(-3) - 3$$

$$A = 0$$

Substituting A and B values in equation (4), we obtain

$$i(t) = 3 - 3e^{-5t}$$

$$\therefore \boxed{i(t) = 3(1 - e^{-5t}) \text{ A}}$$